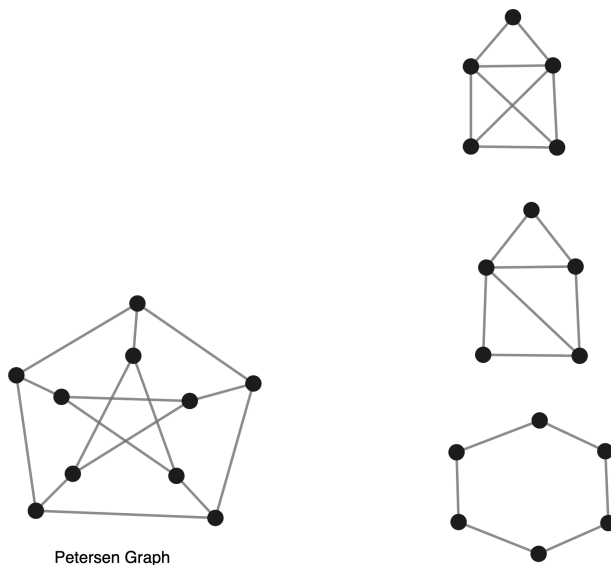


1. For a connected graph $G = (V, E)$ we define the diameter of G to be the maximum distance between any two vertices in V , where the distance between vertices is the minimum number of edges in a path connecting them.
 - (a) Find a formula for the diameter of the complete graph K_n for every n .
 - (b) Find a formula for the diameter of the hypercube graph H_n for every n .

Solution: For part (a) we have that any two vertices are connected to each other by an edge and hence the maximum distance between any two vertices is 1, i.e. the diameter of K_n is 1 for every n . For part (b) we have that an edge on a hypercube graph connects two vertices which differ in exactly one bit, and hence the distance between any two vertices is equal to the number of bits they differ by. Two vertices in H_n can differ by at most n bits and hence the diameter of H_n is n .

2. Determine both the minimum and maximum number of leaves in a spanning tree for the following graphs



Solution: For the peterson graph we have a minimum number of leaves to be 3 and a maximum to be 6. For the other three in descending order we have 2 and 4, 2 and 4, and 2 and 2 respectively for minimum and maximum.

3. Prove or disprove that if $G = (V, E)$ is a connected graph with $|E| \geq |V|$ then G contains a cycle.

Solution: Prove so by induction. It is clear that it is true for $|V| = 3$. Suppose that for a given $|V| = n$ we have that this property holds for all graphs with $|V'| < n$. If the minimal degree of a vertex v in G is 1, we know that this vertex may not be included in a cycle in $G \setminus v$, and hence $G \setminus v$ contains a cycle by induction and so does G . If the minimal degree is 2 then we may consider a maximal path and find an edge connecting its starting point to another point in it creating a cycle.

4. Prove that if a graph $G = (V, E)$ is a tree, then for any edge $e \in E$, we have that $G' = (V, E \setminus \{e\})$ is disconnected.

Solution: Suppose that G is a tree containing an edge $e = v_1v_2$. Then, v_1ev_2 is a path connecting v_1 to v_2 . Suppose by way of contradiction that $G' = (V, E \setminus \{e\})$ is connected, and so there exists a path γ from v_1 to v_2 . Hence, we have that $v_1\gamma v_2ev_1$ is a cycle in G , contradicting that G is a tree. Thus, we must have that G' is disconnected.